

CSE 4128: Image Processing and Computer Vision Laboratory

Assignment No:04

**Assignment Name: Region Descriptor, Feature Matching and GLCM
Texture Analysis**

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Lab4 Tasks

1. Take one of each shape as train image and the other as test image.
2. For each binary image:
 1. Find the Boundary of the object in the image.
 1. Apply erosion to the binary image.
 2. Subtract the eroded image from the binary image to get the border of the region.
 2. Calculate Perimeter, Area, Max Diameter of the region.
 3. Use Area, Perimeter, and Max Diameter to calculate Form factor, Roundness, and Compactness and save it in a txt file.
3. Matching the descriptors of the test and train images to find close Match.

Lab Tasks Output:

```
Desktop/ Image/ Image Processing/ Lab 7/ Lab Task 7

=== Shape Descriptors ===
+-----+-----+-----+
|      | Compactness | Form Factor | Roundness |
+-----+-----+-----+
| c1.jpg | 412.481 | 0.0304653 | 0.913927 |
+-----+-----+-----+
| t1.jpg | 404.876 | 0.0310376 | 0.586762 |
+-----+-----+-----+
| p1.png | 21.2999 | 0.589974 | 0.898139 |
+-----+-----+-----+
| c2.jpg | 428.568 | 0.0293218 | 0.989764 |
+-----+-----+-----+
| t2.jpg | 425.866 | 0.0295078 | 0.582954 |
+-----+-----+-----+
| p2.png | 37.3981 | 0.336016 | 1.01707 |
+-----+-----+-----+

=== Distance Matrix ===
+-----+-----+-----+
|      | c1.jpg | t1.jpg | p1.png |
+-----+-----+-----+
| c2.jpg | 16.0872 | 23.6952 | 407.269 |
+-----+-----+-----+
| t2.jpg | 13.3888 | 20.9895 | 404.566 |
+-----+-----+-----+
| p2.png | 375.083 | 367.479 | 16.1007 |
+-----+-----+-----+

In [4]: |
```

Figure 1: Shape Descriptors and Distance Matrix

- **Assignment 1**

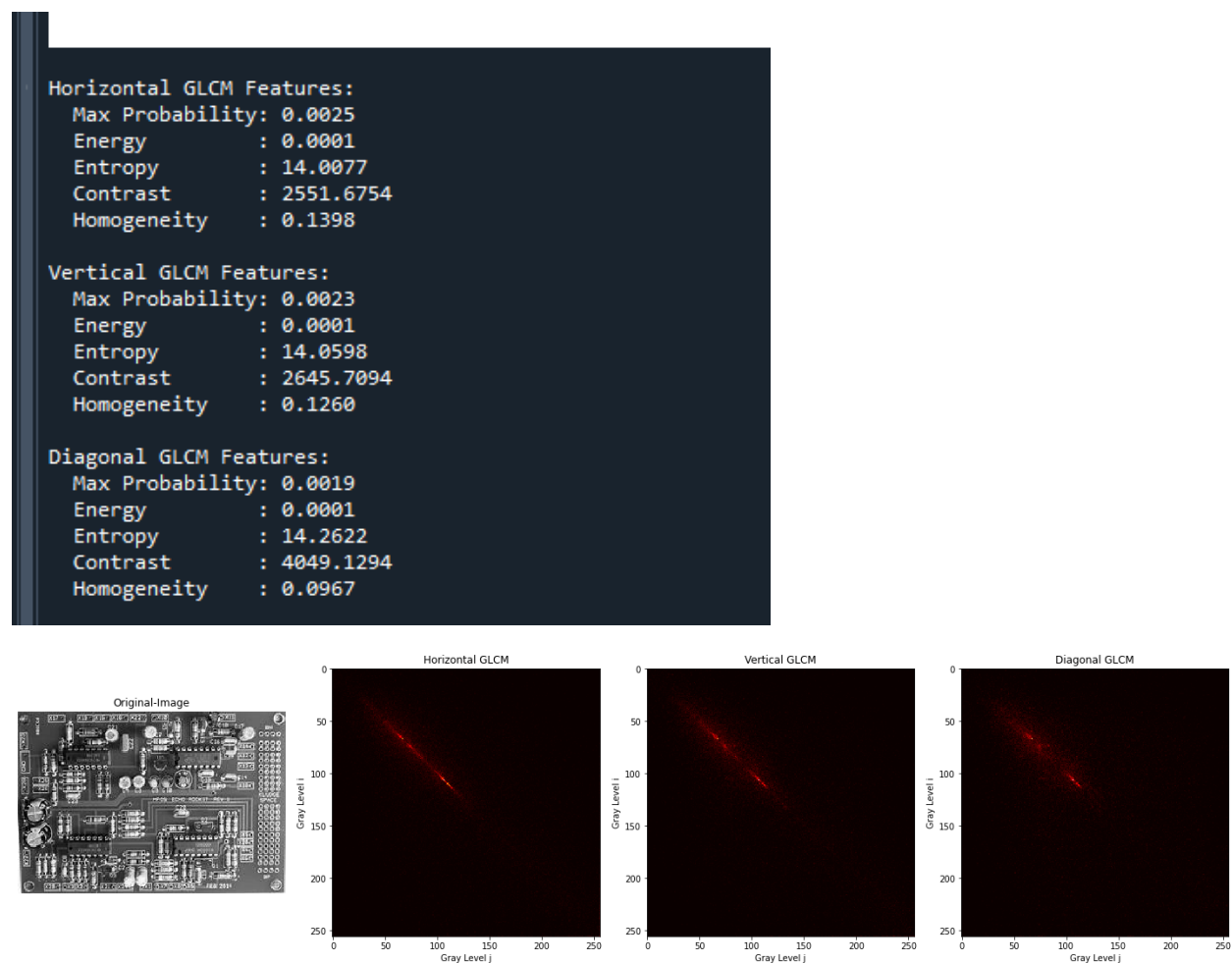
1. Take a gray scale input image and Calculate the

- i) Diagonal, Vertical and horizontal GLCM matrices and visualize them.
- (ii) For each of the 3 GLCM matrices find the 5 texture features

- a) Maximum probability
- b) Energy
- c) Entropy
- d) Contrast
- e) Homogeneity

2. Take a small 5x5 image patch from any of your input image. Apply the above mention operations and demonstrate in your report

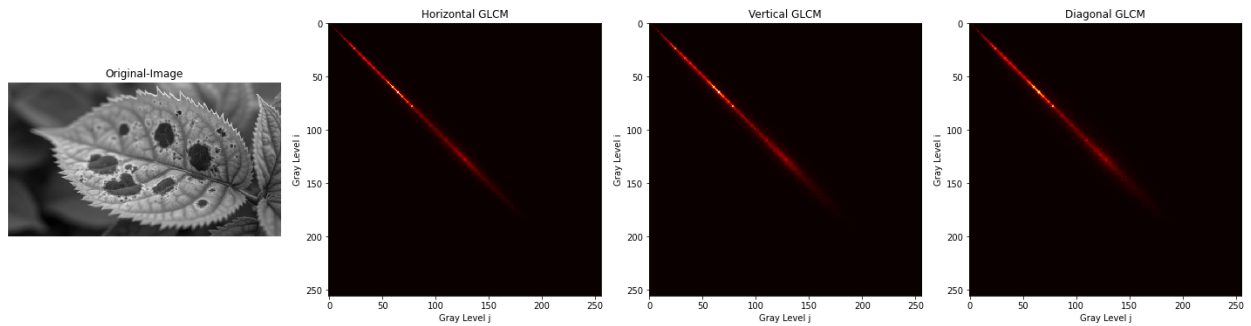
Solutions



```
Horizontal GLCM Features:
  Max Probability: 0.0048
  Energy          : 0.0008
  Entropy         : 11.4757
  Contrast        : 60.6860
  Homogeneity     : 0.4384

Vertical GLCM Features:
  Max Probability: 0.0039
  Energy          : 0.0006
  Entropy         : 11.6826
  Contrast        : 71.8740
  Homogeneity     : 0.3999

Diagonal GLCM Features:
  Max Probability: 0.0036
  Energy          : 0.0005
  Entropy         : 11.9475
  Contrast        : 105.1399
  Homogeneity     : 0.3676
```



Horizontal GLCM Features:

Max Probability: 0.0012

Energy : 0.0004

Entropy : 11.9393

Contrast : 113.1831

Homogeneity : 0.2200

Vertical GLCM Features:

Max Probability: 0.0006

Energy : 0.0002

Entropy : 12.9525

Contrast : 781.4185

Homogeneity : 0.0958

Diagonal GLCM Features:

Max Probability: 0.0006

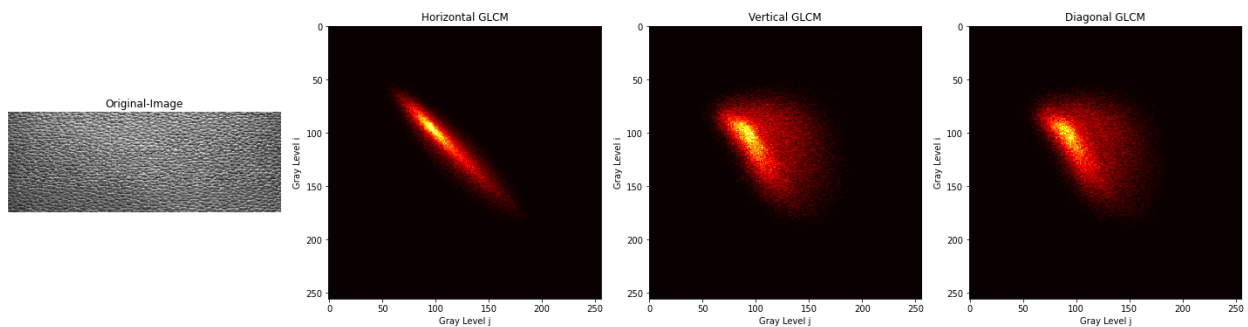
Energy : 0.0002

Entropy : 12.9425

Contrast : 759.2566

Homogeneity : 0.0980

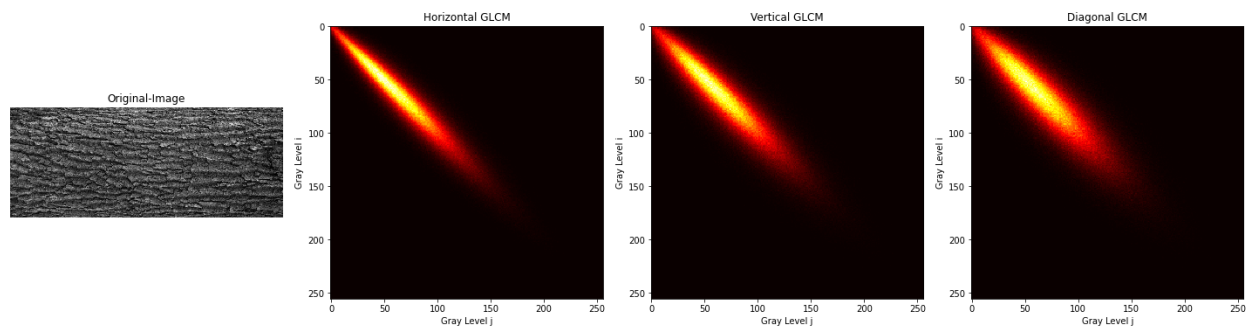
0  Horizontal GLCM



```
Console I/A x
Entropy      : 12.6649
Contrast     : 141.8095
Homogeneity  : 0.2039

Vertical GLCM Features:
Max Probability: 0.0004
Energy        : 0.0002
Entropy       : 13.0233
Contrast      : 233.6090
Homogeneity   : 0.1710

Diagonal GLCM Features:
Max Probability: 0.0003
Energy        : 0.0001
Entropy       : 13.2409
Contrast      : 320.4991
Homogeneity   : 0.1513
```



Assignment 2

2. Take a small 5x5 image patch from any of your input image. Apply the above mention operations and demonstrate in your report

```

In [3]: runfile('C:/Users/USER/Desktop/image/Image-Processing/lab4/assignment/2.py',
Original Patch:
[[74 75 75 76 76]
 [73 73 74 75 75]
 [71 71 71 72 73]
 [68 68 69 70 70]
 [68 68 68 69 70]]
Quantized Patch:
[[6 7 7 8 8]
 [5 5 6 7 7]
 [3 3 3 4 5]
 [0 0 1 2 2]
 [0 0 0 1 2]]
Number of Gray Levels: 9

```

